

Guide to Functions for Data Analysis for Programming Project - Emily Eulitz

Hiiiiii guys, here's a list of the functions I've made and how to use them. Hopefully this will make it easier for you guys to use them and understand how they work without having to read through all the code itself. Feel free to add your own functions you make along the way to help others to understand them! All of these functions are on github under Data as far as I'm aware at the time of writing, and should be accessible from any file.

float getAvgDistance(ArrayList<Flight> flightList)

- Just finds the average distance flown by a given arraylist of flights.
- Takes in:
 1. An arraylist of flights
- Returns:
 - The mean distance flown by all flights in a passed arraylist.

ArrayList<Flight> flightsWhichMatchThisCriterion(String type, String value, ArrayList<Flight> flightList)

- This is the main function used for filtering the data.
- Takes in:
 1. A column type. Possible inputs are:
 - "late"
 - "diverted"
 - "cancelled"
 - "carrier"
 - "orig"
 - "dest"
 - "accDepTimeMinimum"
 - "accDepTimeMaximum"
 - "accArrTimeMinimum"
 - "accArrTimeMaximum"
 - "dateMinimum"
 - "dateMaximum"
 2. A value. Input will depend on the type of data you are trying to search for. All inputs must be in String form. Below are the following possible inputs for each column type (i.e. what the input for "type" is):
 - "late", "diverted", or "cancelled":
 - No input necessary. Simply empty quotation marks ("").
 - "carrier":
 - Any carrier code. e.g "AA" for American Airlines, "UA" for United Airlines, etc.
 - "orig" or "dest":

- Any state abbreviation. E.g. "CA" for California, "AZ" for Arizona, etc.
 - "accDepTimeMinimum", "accDepTimeMaximum", "accArrTimeMinimum", or "accArrTimeMaximum":
 - Any time represented by a simple 4 digit number in the form "TTTT". E.g. 6:03AM = "0603", 5:42PM = "1742", etc.
 - "dateMinimum" or "dateMaximum"
 - A date given in the form "DD". Keep in mind that the whole dataset lies within January 2022, so specifying the month or year is unnecessary. E.g. the 2nd of January = "02", 13th of January = "13", etc.
3. An arraylist of flights.
- Returns:
 - An arraylist of flights which match the criterion specified by the user. E.g. flightsWhichMatchThisCriterion("orig", "AZ", flightlist) will return an arraylist of flights in the arraylist passed which originated in Arizona.

ArrayList<Flight> inThisTimeRange(String type, String min, String max, ArrayList<Flight> flightList)

- One of the main functions used for filtering data based on certain timeframes. This one makes use of flightsWhichMatchThisCriterion() to find a list of flights which are within a specific time or date period.
- Takes in:
 1. A column type. Possible inputs are:
 - "depTimes"
 - "arrTimes"
 - "dates"
 2. A value for the minimum end of a range of values you are trying to find. Below are the possible values of inputs for each column type:
 - "depTimes" or "arrTimes":
 - Any time represented by a simple 4 digit number in the form "TTTT". E.g. 6:03AM = "0603", 5:42PM = "1742", etc.
 - "dates"
 - A date given in the form "DD". Keep in mind that the whole dataset lies within January 2022, so specifying the month or year is unnecessary. E.g. the 2nd of January = "02", 13th of January = "13", etc.
 3. A value for the maximum end of a range of values you are trying to find. Below are the possible values of inputs for each column type:
 - "depTimes" or "arrTimes":
 - Any time represented by a simple 4 digit number in the form "TTTT". E.g. 6:03AM = "0603", 5:42PM = "1742", etc.
 - "dates"
 - A date given in the form "DD". Keep in mind that the whole dataset lies within January 2022, so specifying the month or

year is unnecessary. E.g. the 2nd of January = "02", 13th of January = "13", etc.

4. An arraylist of flights
- Returns:
 - An arraylist of flights which fall within a range specified by the inputs. E.g. `inThisTimeRange(depTimes, "1330", "1540", flightList)` will return an arraylist of flights in the arraylist passed which departed between 13:30 and 15:40. `inThisTimeRange(dates, "02", "15", flightList)` will return an arraylist of flights in the arraylist passed which departed between the 2nd and 15th of January 2022.

float percentWhichMatchThisCriterion(String type, String value, ArrayList<Flight> flightList)

- A function which uses `flightsWhichMatchThisCriterion()` to find the percentage of a particular arraylist that matches a criterion set by the inputs.
- Takes in: (arguments are the same as `flightsWhichMatchThisCriterion()`, just repeating them here for reference)
 1. A column type. Possible inputs are:
 - "late"
 - "diverted"
 - "cancelled"
 - "carrier"
 - "orig"
 - "dest"
 - "accDepTimeMinimum"
 - "accDepTimeMaximum"
 - "accArrTimeMinimum"
 - "accArrTimeMaximum"
 - "dateMinimum"
 - "dateMaximum"
 2. A value. Input will depend on the type of data you are trying to search for. All inputs must be in String form. Below are the following possible inputs for each column type (i.e. what the input for "type" is):
 - "late", "diverted", or "cancelled":
 - No input necessary. Simply empty quotation marks ("").
 - "carrier":
 - Any carrier code. e.g "AA" for American Airlines, "UA" for United Airlines, etc.
 - "orig" or "dest":
 - Any state abbreviation. E.g. "CA" for California, "AZ" for Arizona, etc.
 - "accDepTimeMinimum", "accDepTimeMaximum", "accArrTimeMinimum", or "accArrTimeMaximum":
 - Any time represented by a simple 4 digit number in the form "TTTT". E.g. 6:03AM = "0603", 5:42PM = "1742", etc.
 - "dateMinimum" or "dateMaximum"

- A date given in the form "DD". Keep in mind that the whole dataset lies within January 2022, so specifying the month or year is unnecessary. E.g. the 2nd of January = "02", 13th of January = "13", etc.
3. An arraylist of flights
- Returns:
 - The percentage of flights in a passed array which meet the criterion set by the inputs. The percentage is given in the form of a float and is rounded to two decimal places. E.g. percentWhichMatchThisCriterion("carrier", "AA", flightList) returns the percentage of flights in the passed arraylist which had the carrier American Airlines. The percentage is given in the form "DD.DD", E.g. if 25.42% of flights match this criterion, then "25.42" will be returned (in float form).

percentInThisTimeRange(String type, String min, String max, ArrayList<Flight> flightList)

- Very similar to percentWhichMatchThisCriterion(), only it's used for date ranges instead.
- Takes in: (arguments are the same as inThisTimeRange(), just repeating them here for reference)
 1. A column type. Possible inputs are:
 - "depTimes"
 - "arrTimes"
 - "dates"
 2. A value for the minimum end of a range of values you are trying to find. Below are the possible values of inputs for each column type:
 - "depTimes" or "arrTimes":
 - Any time represented by a simple 4 digit number in the form "TTTT". E.g. 6:03AM = "0603", 5:42PM = "1742", etc.
 - "dates"
 - A date given in the form "DD". Keep in mind that the whole dataset lies within January 2022, so specifying the month or year is unnecessary. E.g. the 2nd of January = "02", 13th of January = "13", etc.
 3. A value for the maximum end of a range of values you are trying to find. Below are the possible values of inputs for each column type:
 - "depTimes" or "arrTimes":
 - Any time represented by a simple 4 digit number in the form "TTTT". E.g. 6:03AM = "0603", 5:42PM = "1742", etc.
 - "dates"
 - A date given in the form "DD". Keep in mind that the whole dataset lies within January 2022, so specifying the month or year is unnecessary. E.g. the 2nd of January = "02", 13th of January = "13", etc.
 4. An arraylist of flights

- Returns:
 - The percentage of flights in a passed array which meet the fall within a certain timeframe set by the inputs. The percentage is given in the form of a float and is rounded to two decimal places. E.g. `percentInThisTimeRange("dates", "02", "18", flightList)` returns the percentage of flights in the passed arraylist which fell between the 2nd and the 18th of January 2022. The percentage is given in the form "DD.DD", E.g. if 25.42% of flights fall within this time, then "25.42" will be returned (in float form).

float roundToTwoDecimals(float numb)

- Just a simple rounding function I found myself using a lot.
- Takes in:
 - A float
- Returns:
 - The passed float rounded to two decimal places

String mostPopValue(String type, ArrayList<Flight> flightList)

- Finds the most popular value of a particular column in a passed arraylist.
- Takes in:
 1. A column type. Possible inputs are:
 - "dest"
 - "orig"
 - "carrier"
 2. An arraylist of flights.
- Returns:
 - The most common value for a given column in a passed arraylist in the form of a String. E.g. `mostPopValue(dest, flightList)` would return the most common destination in the passed arraylist. For "dest" and "orig", the String returned will be the state abbreviation of the most common state (e.g. "AZ" if the most common state is Arizona), and for "carrier" the String returned will be the carrier code of the most common carrier (e.g. "AA" if the most common carrier is American Airlines).

ArrayList<String> stateCodes(ArrayList<Flight> flights)

- Creates an arraylist of all state abbreviations which occur in the passed arraylist.
- Takes in:
 1. An arraylist of flights
- Returns:
 - An arraylist of Strings of all state abbreviations present in the passed arraylist.

ArrayList<String> carrierCodes(ArrayList<Flight> flights)

- Creates an arraylist of all carrier codes which occur in the passed arraylist.
- Takes in:
 - 1. An arraylist of flights
- Returns:
 - An arraylist of all carrier codes present in the passed arraylist.